

Sets

Capital letters denote a set $(A, B, C, \dots)$

If A consists of objects a_1, a_2, a_3 , we say $A = \{a_1, a_2, a_3\}$

S denotes the universal set (contains all objects)

$\emptyset$ denotes the empty set ($\emptyset = \{\}$)

A is a subset of B if everything in A is in B ($A \subset B$) (We also allow equality with $\subset$)

$\emptyset \subset$ of any set

The union of A and B is the set of all objects in A OR B ($A \cup B$)

$A = \{1, 2, 3\}, B = \{4, 5\}, A \cup B = \{1, 2, 3, 4, 5\}$



The intersection of A and B is the set of all objects in A AND B ($A \cap B$)

$A = \{1, 2, 3\}, B = \{1, 2\}, A \cap B = \{1, 2\}$



The complement of A is the set of all objects in S that are not in A (A^c)



$A \cup A^c = S, A \cap A^c = \emptyset$ A and A^c are disjoint

A and B are disjoint / mutually exclusive if they have no object in common ($A \cap B = \emptyset$)



$\emptyset$ and any set are disjoint

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

DeMorgan's Laws

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c$$

$$\left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c$$

Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Nondecreasing and Nonincreasing Sets

Can be equal, increasing can't

A sequence of sets $\{C_n\}$ is nondecreasing if $C_n \subset C_{n+1}$ for all n , so $\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n = \text{Largest}$

A sequence of sets $\{C_n\}$ is nonincreasing if $C_{n+1} \subset C_n$ for all n , so $\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n = \text{Smallest}$



Note

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

Example

Show that the sequence of sets $C_k = \{x: \frac{1}{k} \leq x \leq 3 - \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$

is increasing and find $\lim_{k \rightarrow \infty} C_k$

We know $\frac{1}{k-1} < \frac{1}{k}$, so $3 - \frac{1}{k} < 3 - \frac{1}{k+1}$

Lower bound tends to 0 (decreasing)

Upper bound tends to 3 (increasing)

Therefore, the interval is growing / increasing

$$\lim_{k \rightarrow \infty} C_k = (0, 3)$$

Example

Show that the sequence of sets $C_k = \{x: 2 < x < 2 + \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$

is decreasing and find $\lim_{k \rightarrow \infty} C_k$

We know the lower bound is fixed

$2 + \frac{1}{k}$ tends to 2 (decreasing)

Therefore, the interval is decreasing

$$\lim_{k \rightarrow \infty} C_k = [2, 2) = \emptyset$$